MODULE 18 · DOMAIN 2 - ML MODEL AND FM DEVELOPMENT

Evaluating FMs and RAG

BLEU, ROUGE, BERTScore, LLM-as-a-judge, and the two halves of a RAG system.

MODULE 18 of 43 DOMAIN 2 - Model Dev WEIGHT 24%

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There are **many good answers** to most questions, and the metric has to tolerate that.

What makes FM evaluation different

- **No single correct answer.** Many outputs can be equally good.
- **Quality is multi-dimensional:** correct, complete, faithful, appropriately toned, safe.
- **Non-determinism.** The same prompt can produce different output, from module 16.
- **Reference answers are expensive**, and sometimes nobody can write one.
- **New failure modes:** confident fabrication, and refusing a reasonable request.

> ON THE EXAM
>
> Hallucination has no equivalent in classical ML. A tabular model cannot invent a plausible customer who does not exist.

Three approaches, and what each costs

APPROACH	WHAT IT GIVES YOU
Automatic	Traditional NLP metrics and exact string matching. Fast and cheap
LLM-as-a-judge	Human-like judgement at scale, for a fraction of the cost and time
Human-based	Real human rating. Slowest, dearest, and the reference standard

ON THE EXAM

AWS is blunt about the first row: traditional metrics 'did not provide a strong correlation with human evaluators'.

What an evaluation job needs

INPUT	WHAT YOU SUPPLY
Prompt dataset	JSONL in S3. One prompt per line, with optional reference and context
Model or system	A Bedrock model, a knowledge base, or your own responses via BYOI
Metrics	Which built-in or custom metrics to score
Judge model	Which model performs the scoring, for LLM-as-a-judge jobs
Output location	An S3 path for results, plus IAM permissions to write there

ON THE EXAM

The prompt dataset is the only part that takes real effort. Everything else is configuration.

Built-in task types

- Automatic evaluation offers preset task types with curated datasets and metrics.
- **Text summarisation, question answering, text classification, open-ended generation.**
- Each comes with appropriate metrics already selected, so you do not choose blind.
- Useful as a **first screen** across many models, before a judge job on the shortlist.
- Curated datasets are generic. They tell you about the model, not about your use case.

> ON THE EXAM
>
> Two stages: screen broadly with automatic evaluation, then judge the shortlist on your own prompts.

BLEU, ROUGE and BERTScore all ask the same thing: **how close is this to the reference?**

BLEU

- Compares **n-grams** in the output against n-grams in the reference. Precision-oriented.
- Built for **machine translation**, where the output should closely match a known correct translation.
- Includes a brevity penalty, so a short output cannot score well by saying little.
- It rewards using the reference's exact words. A correct paraphrase scores badly.
- It does not understand meaning at all. It is string overlap with arithmetic on top.

> ON THE EXAM
>
> BLEU asks: how much of what I said appears in the reference? That suits translation and suits almost nothing else well.

ROUGE

- Also n-gram overlap, but **recall-oriented**: how much of the reference did the output cover?
- Built for **summarisation**, where covering the key content matters more than exact wording.
- **ROUGE-N** for n-gram overlap. **ROUGE-L** for the longest common subsequence.
- Same fundamental limit: it counts word overlap, not meaning.
- A summary can score well by copying phrases without being a good summary.

> ON THE EXAM
>
> BLEU is precision, ROUGE is recall. That single contrast is the most likely thing to be tested about either.

BLEU or ROUGE?

BLEU

Precision-oriented

- Of what I said, how much matched?
- Designed for machine translation.
- Penalises adding extra content.
- Includes a brevity penalty.

ROUGE

Recall-oriented

- Of the reference, how much did I cover?
- Designed for summarisation.
- Penalises leaving content out.
- ROUGE-L uses longest common subsequence.

THE TIEBREAKER

Translation is BLEU. Summarisation is ROUGE. Both need a reference answer, and both are blind to paraphrase.

BERTScore and semantic similarity

- Compares **embeddings** of the output and the reference, not their words.
- So a correct paraphrase scores well, which is exactly what BLEU and ROUGE get wrong.
- This is module 09's embedding machinery, applied to evaluation.
- It still needs a reference answer. It only removes the exact-wording requirement.
- It can score a fluent, on-topic, factually wrong answer highly. Similar is not correct.

> ON THE EXAM
>
> BERTScore fixes the paraphrase problem and not the truth problem. Two statements can be semantically close and one of them false.

Other automatic metrics

METRIC	WHAT IT DOES
Exact match	Output equals the reference exactly. Only useful for short, closed answers
F1 on tokens	Token-level precision and recall. Common in question answering
Perplexity	How surprised the model is by the text. A model-fluency measure, not a quality one
Semantic similarity	Embedding distance between output and reference

ON THE EXAM

Exact match is the right metric more often than people expect: extraction, classification and closed questions all have one right answer.

Other evaluation tooling

- **fmeval** is the AWS open-source foundation model evaluation library.
- **Ragas** and **DeepEval** are widely used open-source frameworks for RAG evaluation.
- They cover similar ground, with more control and more work to operate.
- Managed MLflow from module 19 can track evaluation results over time.
- For the exam, answer with **Bedrock Evaluations** unless a scenario rules it out.

> ON THE EXAM
>
> Same pattern as the rest of the course: know the alternatives exist, and answer with the AWS-native service the guide names.

If no reference exists, use a model to judge the answer instead of matching it.

LLM-as-a-judge

- A **judge model** scores each response against criteria you select, and you choose the judge.
- AWS: human-like evaluation quality at **a fraction of the cost and time** of human review.
- **Scores are normalised 0 to 1**, and every score comes with a natural language explanation.
- No reference answer needed for most metrics, which is the point of it.
- Preview December 2024, generally available 20 March 2025.

> ON THE EXAM
>
> The natural language explanation is the underrated part. A score of 0.4 with a sentence saying why is actionable; a bare 0.4 is not.

What a judge can score

METRIC	THE QUESTION IT ASKS
Correctness	Is the answer factually right?
Completeness	Does it cover everything the question asked?
Faithfulness	Is it supported by the provided context? Hallucination detection
Professional style and tone	Does it sound the way it should?
Harmfulness	Is the content harmful?
Answer refusal	Did it decline a request it should have answered?
Stereotyping	Does it reproduce stereotypes?

ON THE EXAM

The last three are responsible AI metrics. Answer refusal is the one people forget: over-refusing is a real failure, not caution.

Responsible AI metrics in practice

- **Harmfulness:** does the output contain content that could cause harm?
- **Stereotyping:** does it reproduce stereotypes about groups of people?
- **Answer refusal cuts both ways.** Refusing a reasonable request is a failure too.
- An over-cautious model that declines ordinary questions is unusable, and scores well on harmfulness.
- Evaluate these **alongside** quality metrics, not instead of them. Safe and useless is not success.

> ON THE EXAM
>
> Safety and usefulness trade against each other. Measuring only one of them produces a system that is excellent at the other's expense.

Where LLM-as-a-judge falls down

- The judge has **biases of its own**: it can prefer longer answers, or its own writing style.
- A judge from the same family as the model being judged may favour it.
- It struggles on **specialist domains** where it lacks the knowledge to judge correctness.
- **Calibrate it against human ratings** on a sample, at least once, before trusting it.
- It is a very good proxy for human judgement. It is not a replacement for ever asking a human.

> ON THE EXAM
>
> The failure mode is agreement without accuracy: a judge confidently scoring specialist answers it is not qualified to assess.

Reliability of judge scores

- A judge is a model, so its scores carry **variance**. Run the same job twice and they differ slightly.
- Small differences between candidates may be noise, exactly as in module 17.
- **Set the judge's temperature low**, so scoring is as repeatable as it can be.
- Read the natural language explanations on a sample. They reveal when a judge misunderstood the task.
- Averaging over more prompts reduces noise. Thirty prompts is a noisy estimate of anything.

> ON THE EXAM
>
> Module 17's warning applies here too: choosing the model with the higher third decimal place is choosing randomly.

Human evaluation

- Routes responses to a **work team** for subjective rating, on criteria you define.
- The team can be your own employees, or an AWS-managed workforce.
- It is orchestrated through **SageMaker Ground Truth work teams**, from module 07.
- Ground Truth is closed to new customers, so this path may be unavailable to you.
- Use it to **calibrate**, on a sample, rather than to evaluate everything.

> ON THE EXAM
>
> Human evaluation is slow and expensive, so spend it where it buys the most: checking that your cheaper method agrees with people.

A RAG system has **two things that can fail**, and one number cannot tell you which did.

Retrieve only, or retrieve and generate?

RETRIEVE ONLY

Isolates the retriever

- Evaluates the retrieval stage alone.
- Metrics: context relevance, context coverage.
- Answers: did we fetch the right passages?
- Use it to tune chunking and embeddings.

RETRIEVE AND GENERATE

Evaluates end to end

- Retrieval plus the generated answer.
- Correctness, completeness, faithfulness.
- Answers: is the final response good?
- Use it to compare generating models.

THE TIEBREAKER

Diagnose with retrieve-only, judge with retrieve-and-generate. Both are powered by LLM-as-a-judge underneath.

Context relevance and context coverage

- **Context relevance:** are the retrieved passages actually about the question?
- **Context coverage:** do they contain everything needed to answer it?
- **They fail differently.** Low relevance is noise; low coverage is a gap.
- Low relevance points at the embedding model, the query, or the index.
- Low coverage points at **chunking**, or at top-k being too small. Module 10.

> ON THE EXAM
>
> This is the pair that turns 'retrieval is bad' into an actionable diagnosis, because the two have different fixes.

Faithfulness, or hallucination detection

- Asks whether the answer is **supported by the retrieved context**.
- An answer can be **correct and unfaithful**: true, but not from the documents you gave it.
- And **faithful but wrong**, if the retrieved documents were themselves wrong.
- In a grounded system, **unfaithful is a failure even when the answer is true**.
- This is the metric a regulated or citation-bearing application cares about most.

> ON THE EXAM
>
> Correctness and faithfulness are different questions. A system that must cite its sources needs the answer to come from them.

Citation coverage and citation precision

- **Citation coverage:** is the content of the answer actually backed by the citations given?
- **Citation precision:** are the citations given genuinely relevant to what was said?
- Both were added at general availability in March 2025.
- A citation that looks authoritative and does not support the claim is worse than no citation.
- Module 01 called citation the giveaway for choosing retrieval. This is how you check it works.

> ON THE EXAM
>
> Users trust citations more than answers. A wrong answer with a plausible source attached is the most dangerous output a RAG system can produce.

Bring your own inference responses

- **BYOI** lets you supply responses your own system produced, in the required input format.
- So you can evaluate **any model or RAG system hosted anywhere**, including on-premises.
- The evaluation machinery and the metrics are the same. Only the source of responses changes.
- Useful for comparing a Bedrock system against something you already run.
- Added alongside RAG evaluation at general availability.

> ON THE EXAM
>
> This turns Bedrock Evaluations from a Bedrock feature into an evaluation service you can point at anything.

Custom metrics

- You can define your own metric in addition to the built-in ones.
- The example AWS gives: adherence to **your specific brand voice**.
- Also useful for domain rules: did it include the required disclaimer, did it avoid advice?
- Defined as criteria the judge model applies, so it needs a clear, judgeable definition.
- A vague custom metric produces vague scores. Precision in the definition is the work.

> ON THE EXAM
>
> Anything a human reviewer could consistently judge from a written rule can usually become a custom metric.

Building an evaluation dataset

- **Real questions** from users or logs, not questions invented in a workshop.
- Include the **hard cases deliberately**: ambiguous, multi-part, and out-of-scope questions.
- Include questions the system **should refuse**, to test answer refusal in both directions.
- Thirty to fifty is enough to discriminate. A few hundred is comfortable.
- **Freeze it.** A changing evaluation set makes runs incomparable, from module 15.

> ON THE EXAM
>
> The dataset is the asset, not the tooling. Bedrock Evaluations is easy to adopt; a good evaluation set takes real effort.

Maintaining the evaluation set

- **Add every production failure** to it. A bug found once should never recur unnoticed.
- That makes the set grow in exactly the directions your system is weak.
- Adding cases changes the baseline, so record which version of the set produced which score.
- Keep a frozen core for comparability, and a growing extension for coverage.
- Human review corrections from module 07 are a natural source of new cases.

> ON THE EXAM
>
> An evaluation set that never changes stops finding new problems. One that changes constantly makes runs incomparable. Keep both.

Evaluation as a quality gate

- Run an evaluation job when a prompt, a model, or the knowledge base configuration changes.
- **Gate the release on the score**, exactly as module 17 gated model registration.
- Compare across evaluation jobs to see whether a change actually improved anything.
- Without this, 'better' means whichever examples somebody happened to look at.
- Module 28 wires this into a pipeline.

> ON THE EXAM
>
> A prompt change is a production change with no deployment, from module 16. A quality gate is what stops it shipping untested.

Comparing evaluation jobs

- Bedrock lets you **compare results across evaluation jobs** directly.
- So you can change **one variable** and see the effect: chunk size, embedding model, reranker, judge.
- That turns RAG tuning from opinion into measurement, which modules 09 and 10 could not do alone.
- Change one thing per run. Two changes and you cannot attribute the difference.
- Keep the prompt dataset frozen across runs, or the comparison is meaningless.

> ON THE EXAM
>
> This is what makes module 10's irreversible chunking decision survivable: you can measure two data sources before committing.

Regression testing a generative system

- Generative systems **degrade quietly**: a model version moves, documents change, a prompt is edited.
- Run the same frozen evaluation set on a schedule, and watch the scores over time.
- A **drop between runs** is the signal. The absolute number matters less than the change.
- Investigate per-metric: a fall in faithfulness and a fall in relevance mean different things.
- This is module 29's drift monitoring, applied to a system with no accuracy metric.

> ON THE EXAM
>
> Without a repeated evaluation, the first sign of degradation is a user complaint, and by then it has been happening for weeks.

Evaluating agents is harder again

- An agent chooses its own path, so two runs of the same task can differ legitimately.
- You evaluate the **outcome**, and separately whether the **steps taken** were sensible.
- New failures: wrong tool chosen, tool called with bad arguments, loops, giving up early.
- Cost becomes part of quality, because an agent that succeeds in forty calls is not a success.
- Module 27 builds agents and module 30 monitors them. This is the evaluation half.

ON THE EXAM

For a model you evaluate one output. For an agent you evaluate a trajectory, and trajectories vary even when they are all correct.

What evaluation costs

APPROACH	COST SHAPE
Automatic metrics	Cheapest. Arithmetic on text, no model calls
LLM-as-a-judge	One or more judge model calls per test case
RAG evaluation	Retrieval, generation, and judging per case. The dearest of the three
Human evaluation	Human time. Slowest and most expensive by a wide margin
Practical scale	Roughly 20 to 30 minutes for 100 prompts on a RAG job

ON THE EXAM

RAG evaluation runs several model calls per question. A 500-question set is a real cost, which is an argument for a smaller, better set.

Evaluate quality, latency and cost together

- The highest-scoring model is often the slowest and the dearest.
- Record **tokens used and response time** alongside the quality score for every candidate.
- Then ask what the quality difference is **worth** at your request volume.
- A two percent quality gain that triples cost per request is rarely the right trade.
- Module 11's distillation exists for exactly this: keep the quality, drop the cost.

> ON THE EXAM
>
> The cheapest model that clears the bar is usually the right answer. Evaluation should tell you where the bar is, not who wins outright.

Symptoms, and what they usually mean

SYMPTOM	USUAL CAUSE
Good answers scoring badly on ROUGE	Correct paraphrase. Use BERTScore or a judge
High similarity score, answer is wrong	Semantic closeness is not truth. Add correctness
Answer is right but not in the documents	Unfaithful. The model used training knowledge
Retrieved passages on topic, answer incomplete	Low context coverage. Chunking or top-k
Retrieved passages off topic	Low context relevance. Embedding, query or index
Citations look authoritative, do not support the claim	Low citation precision
Judge scores disagree with your reviewers	Judge not calibrated, or the domain is too specialist

ON THE EXAM

Rows four and five are the diagnostic pair. Same complaint from a user, two different fixes, and only a retrieve-only job separates them.

Measure the **stage that can fail**, not the system as a whole.

A team builds a service that summarises long customer complaint emails into three-sentence briefs for managers. They want to compare four foundation models before choosing one.

Nobody has written reference summaries, and the team estimates it would take weeks to produce enough of them. Managers care about whether the brief captures the complaint accurately and reads professionally.

How should they evaluate the four models?

A ROUGE against summaries generated by the largest of the four models

B BLEU, since it includes a brevity penalty suited to three-sentence output

C LLM-as-a-judge on correctness, completeness and professional style and tone

D Perplexity, as a reference-free measure of output quality

◆ PAUSE AND DECIDE

How should they evaluate the four models?

A ROUGE against summaries generated by the largest of the four models

Treats one candidate's output as ground truth, so it measures similarity to that model rather than quality. It would rank that model highest by construction.

B BLEU, since it includes a brevity penalty suited to three-sentence output

BLEU is precision-oriented and built for translation. It needs a reference, which does not exist, and rewards exact wording over good summarising.

C LLM-as-a-judge on correctness, completeness and professional style and tone

No reference needed, and the three named metrics map directly onto what managers said they cared about. Scores come with explanations.

D Perplexity, as a reference-free measure of output quality

Reference-free, and it measures fluency rather than quality. A fluent summary that misses the complaint scores well.

THE SIGNAL

'Nobody has written reference summaries' eliminates every classical metric at once. That sentence is doing all the work.

A policy assistant built on a Bedrock knowledge base is reviewed by the compliance team. They find cases where the retrieved passages are clearly the right ones, but the generated answer states a rule that appears nowhere in them.

The answers are often factually correct, because the model knows the general rule. But compliance requires every answer to come from the firm's own documents.

Which evaluation should they run, and on which metric?

A Retrieve and generate, focusing on faithfulness

B Retrieve only, focusing on context relevance

C Retrieve and generate, focusing on correctness

D Model evaluation with BERTScore against the source documents

◆ PAUSE AND DECIDE

Which evaluation should they run, and on which metric?

A Retrieve and generate, focusing on faithfulness

Faithfulness asks whether the answer is supported by the retrieved context. That is exactly the compliance requirement, and it is a generation-stage question.

B Retrieve only, focusing on context relevance

The scenario states retrieval is already fetching the right passages. Measuring the stage that is working will show a good score and change nothing.

C Retrieve and generate, focusing on correctness

Right job type, wrong metric. The answers are already factually correct, so correctness would score well while the compliance failure continues.

D Model evaluation with BERTScore against the source documents

Semantic similarity to a document is not groundedness. A generally-correct answer is semantically close to the right document without deriving from it.

THE SIGNAL

Correct and unfaithful is the exact combination described. In a grounded system that is a failure, even though every answer is true.

> A support knowledge base gives answers that are on the right subject but frequently incomplete, missing a step or a condition that the documentation does contain.
>
> The team has changed the generating model twice with no improvement. They want to establish whether the problem is in retrieval before changing anything else.

What should they run?

A Retrieve and generate, and compare completeness across generating models

B Human evaluation on a sample of answers

C Model evaluation with ROUGE against the documentation

D A retrieve-only job, and look at context coverage

◆ PAUSE AND DECIDE

What should they run?

A. Retrieve and generate, and compare completeness across generating models

Repeats what they already did twice. An end-to-end score cannot say whether the missing content was ever retrieved.

B. Human evaluation on a sample of answers

Confirms the answers are incomplete, which is already known. Slow and expensive, and it still does not isolate the stage.

C. Model evaluation with ROUGE against the documentation

Compares wording to a document rather than assessing a pipeline. It cannot separate retrieval from generation either.

D. A retrieve-only job, and look at context coverage

Isolates retrieval. Low coverage means the passages never contained the missing step, so no generating model could have included it.

THE SIGNAL

Two model changes with no improvement is the clue. When changing the generator does nothing, the generator is probably not the problem.

A team evaluates a customer-service chatbot using BLEU, comparing responses against transcripts of what human agents actually said in similar past conversations.

BLEU scores are low across every candidate model, yet reviewers reading the responses judge them helpful, accurate and well written. The team concludes all four models are unsuitable.

What is wrong with this evaluation?

A The reference transcripts are too short for BLEU's brevity penalty to work

B BLEU measures n-gram overlap with one reference, so valid alternative phrasings score badly

C BLEU requires at least four reference answers per prompt to be reliable

D Nothing is wrong. Low BLEU indicates the models are genuinely unsuitable

◆ PAUSE AND DECIDE

What is wrong with this evaluation?

A. The reference transcripts are too short for BLEU's brevity penalty to work

The brevity penalty stops short outputs scoring well. It is not a length requirement on the reference, and it does not explain the disagreement.

B. BLEU measures n-gram overlap with one reference, so valid alternative phrasings score badly

There are many good ways to answer a customer. BLEU rewards matching one agent's exact words, which is not what quality means here.

C. BLEU requires at least four reference answers per prompt to be reliable

Multiple references do help BLEU, and there is no such threshold. It also would not rescue a metric that is measuring the wrong property.

D. Nothing is wrong. Low BLEU indicates the models are genuinely unsuitable

Human reviewers judged the responses helpful, accurate and well written. When the metric and the humans disagree, the metric is the thing to question.

THE SIGNAL

The disagreement between BLEU and the reviewers is the evidence. A metric that contradicts careful human judgement is measuring the wrong thing.

What to take from this module

- **BLEU is precision and translation. ROUGE is recall and summarisation.** Both need a reference.
- **BERTScore fixes paraphrase, not truth.** Semantic closeness is not correctness.
- **No reference means LLM-as-a-judge.** Scores 0 to 1, with explanations.
- **Correct and unfaithful is still a failure** in a grounded system.
- **Retrieve-only to diagnose, retrieve-and-generate to judge.** Coverage and relevance fail differently.

> ON THE EXAM
>
> Module 19 closes Domain 2 with explainability, experiment tracking and managed MLflow.